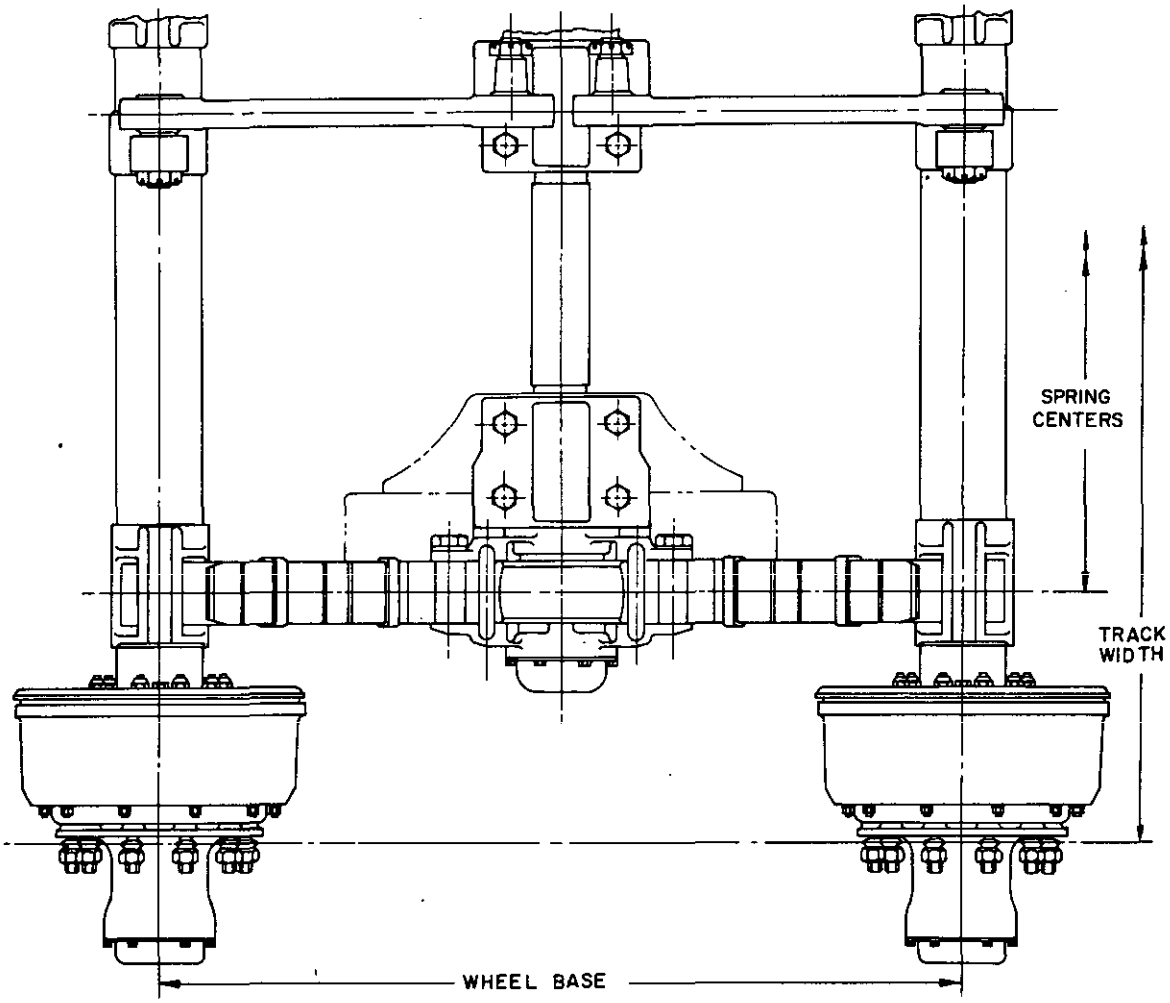
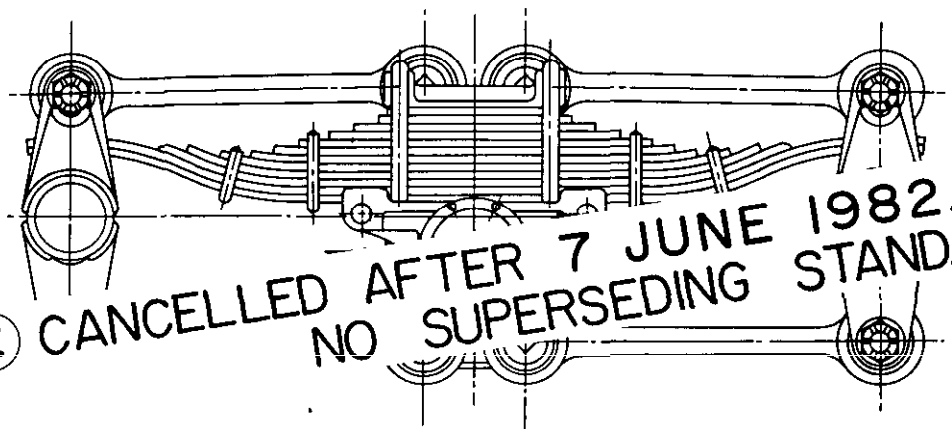
FED. SUP CLASS
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TYPE 'A'

FOR NOTES SEE SHEET 3

(E) CANCELLED AFTER 7 JUNE 1982.
NO SUPERSEDING STANDARD.

APPROVED 27 FEBRUARY 1958 REVISED (A) 12 JULY 1963 (B) 21 DECEMBER 1966 (C) 22 NOV 1972 (D) 20 OCTOBER 1976 (E) 7 JUN 82

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ARMY-AT Other Cust	TITLE AXLE ASSEMBLY, VEHICULAR NON DRIVING, TANDEM FOR TACTICAL WHEELED VEHICLES	MILITARY STANDARD MS 35866
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 1 OF 8

DD FORM 1 SEP 67 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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MS PART NO.	MS 35866-A1	MS 35866-A2	MS 35866-A3	MS 35866-A4	MS 35866-A5
SUSPENSION INSTALLATION (TANDEM)	—	8758757	7979405	10950340	11603095
LOAD BEARING CAPACITY (LBS)	36000 LBS	20000 LBS	36000 LBS	36000 LBS	36000 LBS
BOGIE ASSEMBLY	8710734	8758757	7979405	10950340	11603075
FRONT AXLE	8710727	8758911	7979408	10950341	11603076
REAR AXLE	8710727	8758911	7979404	10950341	11603076
TRUNNION ASSEMBLY (PIVOT AXLE)	7979355	7397986	7979355	7979355	11603077
BRAKE ASSEMBLY LH.	8710719	8336701	7979408 (REF)	—	8376667
BRAKE ASSEMBLY RH.	8710720	8336702	7979404 (REF)	—	8376667
TORQUE ROD ASSEMBLY	7979189	7521838	7979189	7979189	7979189
SPRINGS (MAIN)	7979217	7368629	7979217	7979217	7979217
SPRINGS (AUXILIARY)	NONE	8758890	NONE	NONE	NONE
TUBE ASSEMBLY FRONT AXLE	8710737	10869564	7979344	10950323	11591870
TUBE ASSEMBLY REAR AXLE	8710737	10869564	7979344	10950323	11591870
HUB & DRUM ASSEMBLY L.H.	8710721	—	7979342 (REF)	8710721	8710721
HUB & DRUM ASSEMBLY R.H.	8710722	—	7979343 (REF)	8710722	8710722
HUB	8710723	7263712	7409395 (REF)	8710723	8710723
DRUM	7979233	7411425	7979233	7979233	7979233
ADAPTER	8710724	7413231	7979323	8710724	8710724
WHEEL BEARING ASSEMBLY (OUTER)	MS19081-181	MS19081-112	MS19081-181	MS19081-181	MS19081-181
WHEEL BEARING ASSEMBLY (INNER)	MS19081-182	MS19081-112	MS19081-182	MS19081-182	MS19081-182
RELATED PARTS AND DATA					
WHEELS, P/N	MS 53044-6	MS 53044-5	MS 53044-6	MS 53044-6	11601389
WHEELS, SIZE	20 x 7.5	20 x 7.5	20 x 7.5	20 x 7.5	22.5 X 14
WHEELS, OFFSET	6 11/16	6 3/16	6 11/16	6 11/16	7/16
WHEELS, SINGLE OR DUAL	DUAL	DUAL	DUAL	DUAL	SINGLE
TIRES	MS 35388-21	MS 35388-16	MS 35388-21	MS 35388-21	11603234
TIRES, SIZE	11.00 x 20	9.00 x 20	11.00 x 20	11.00 x 20	18.00 X 22.5
TUBES	MS 35392-15	MS 35392-13	MS 35392-15	MS 35392-15	—
BRAKE SIZE	16 1/2 x 5 1/2	15 x 3 11/32	16 1/2 x 5 1/2	16 1/2 x 5 1/2	16 1/2 X 5 1/2
TRACK WIDTH	72	70	72	72	81 1/4
WHEELBASE	52	48	52	52	52
SPRING CENTERS	39 3/4	40 1/4	39 3/4	39 3/4	49

P.A.
Other Cust

PROCUREMENT SPECIFICATION

SUPERSEDES:

TITLE
AXLE ASSEMBLY, VEHICULAR
NON DRIVING, TANDEM FOR
TACTICAL WHEELED VEHICLES

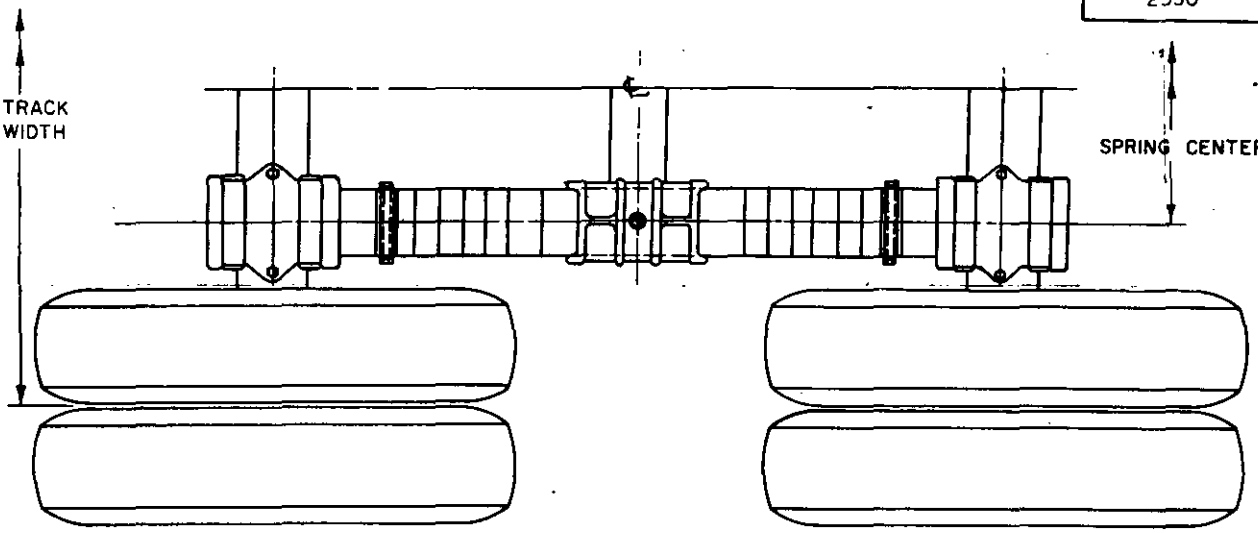
MILITARY STANDARD
MS 35866
SHEET 2 OF 8

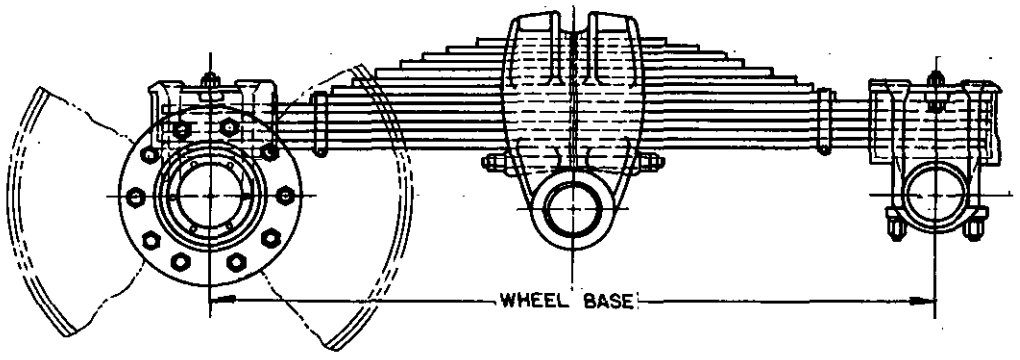
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TYPE "B"

NOTES:

1. LOAD SHALL BE LIMITED BY TIRE, AXLE, OR SPRING CAPACITY.
2. WHEELS AND TIRES ARE NOT PART OF AXLE ASSEMBLY AND ARE SHOWN FOR INFORMATION ONLY.
3. DIMENSIONS SHOWN ARE IN INCHES AND ARE FOR ENGINEERING REFERENCE ONLY.
4. THREADS SHALL BE IN ACCORDANCE WITH SCREW THREAD STANDARDS FOR FEDERAL SERVICES, HANDBOOK H28.
5. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
6. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS, OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
7. MARKING SHALL CONSIST OF THE MS PART NUMBER AND MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.
8. DETAILS OF THIS ASSEMBLY SHALL BE IN ACCORDANCE WITH REFERENCED DRAWINGS, WHICH MAY BE OBTAINED FROM U.S. ARMY TANK-AUTOMOTIVE MATERIEL READINESS COMMAND, WARREN, MICHIGAN 48090.

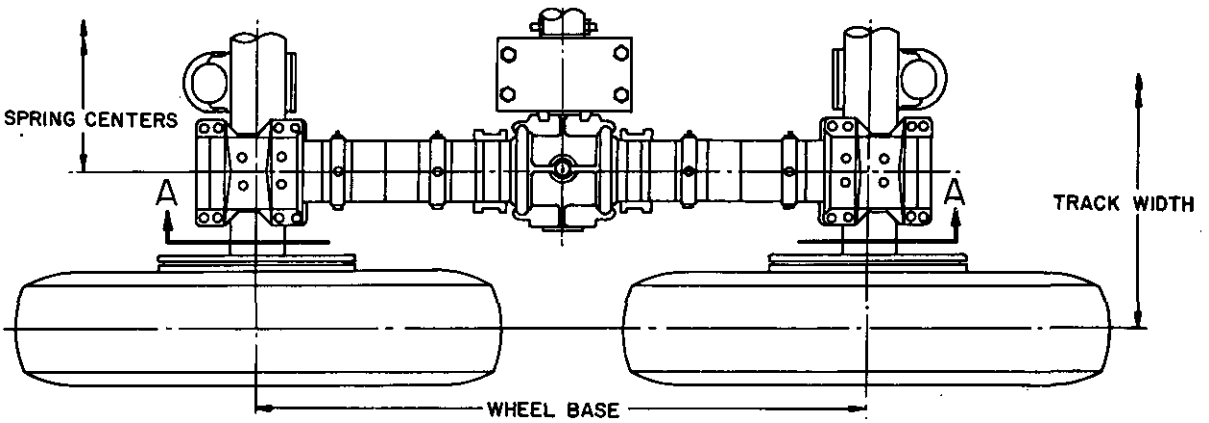
APPROVED 27 FEBRUARY 1958 REVISED

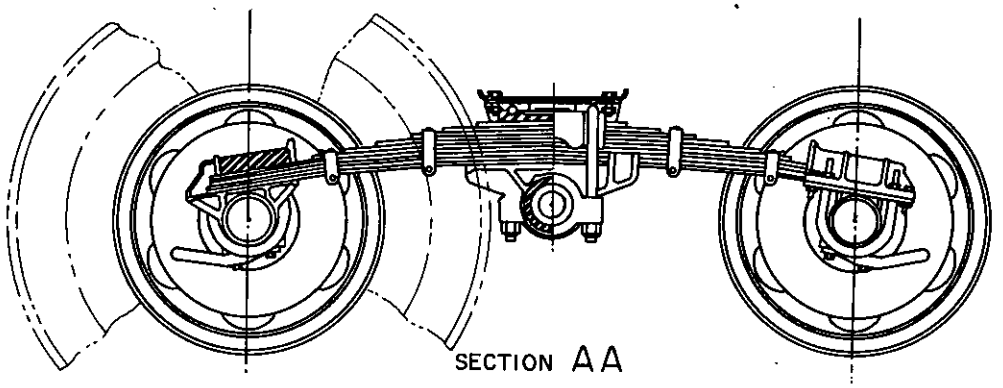
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MS PART NO.	MS 35866-B1	MS 35866-B2
SUSPENSION INSTALLATION (TANDEM)	8703507	8733831
LOAD BEARING CAPACITY	44000 (LBS)	36000 (LBS)
BOGIE ASSEMBLY	8703507	8733831
FRONT AXLE	8703600	8710738
REAR AXLE	8703600	8710739
TRUNION ASSEMBLY (PIVOT AXLE)	8703518	8733751
BRAKE ASSEMBLY L.H.	—	8710719
BRAKE ASSEMBLY R.H.	—	8710720
TORQUE ROD ASSEMBLY	—	—
SPRINGS, MAIN	8703601	8733785
SPRINGS, AUXILIARY	NONE	NONE
TUBE ASSEMBLY FRONT AXLE	8703576	8710702
TUBE ASSEMBLY REAR AXLE	8703576	8710702
HUB & DRUM ASSEMBLY L.H.	—	8710721
HUB & DRUM ASSEMBLY R.H.	—	8710722
HUB	8703603	8710723
DRUM	8703567	7979233
ADAPTER	8703564	8710724
WHEEL BEARING (OUTER)	MS 19081-134	MS19081-181
WHEEL BEARING (INNER)	MS 19081-138	MS19081-182
RELATED PARTS AND DATA		
WHEELS, P/N	MS 53044-8	MS 53044-6
WHEELS, SIZE	20 X 10	20 X 7.5
WHEELS, OFFSET	8 7/8	6 11/16
WHEELS, SINGLE OR DUAL	DUAL	DUAL
TIRES	MS 35388-25	MS 35388-21
TIRES, SIZE	14.00 X 20	11.00 X 20
TUBES	MS 35392-18	MS 35392-15
BRAKE SIZE	16 1/2 X 7 1/2	16 1/2 X 5 1/2
TRACK WIDTH	82	72
WHEEL BASE	52	52
SPRING CENTERS	39 1/4	34

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TYPE 'C'

APPROVED 27 FEBRUARY 1958 REVISED

P.A. Other Cust	TITLE AXLE ASSEMBLY, VEHICULAR NON-DRIVING, TANDEM FOR TACTICAL WHEELED VEHICLES	MILITARY STANDARD MS 35866
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 5 OF 8

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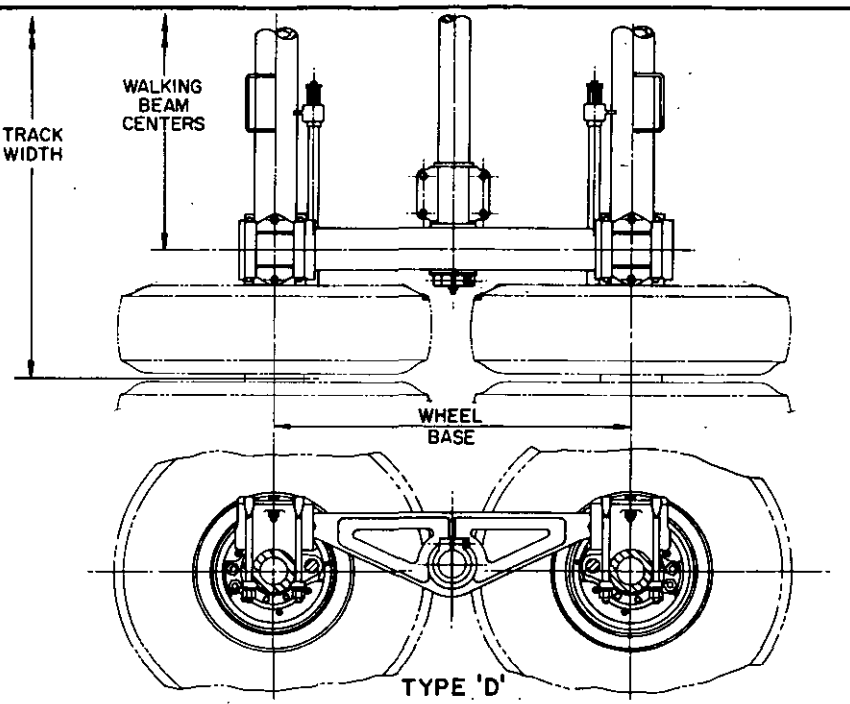
MS PART NO.	MS35866-C1	MS35866-C2
SUSPENSION INSTALLATION (TANDEM)	—	10869680
LOAD BEARING CAPACITY	20000 (LBS)	20000 (LBS)
BOGIE ASSEMBLY	8742430	10869680
FRONT AXLE	8742434	8742434
REAR AXLE	8742434	8742434
TRUNION ASSEMBLY (PIVOT AXLE)	8408989	8408989
BRAKE ASSEMBLY L.H.	8723800	8723800
BRAKE ASSEMBLY R.H.	8723801	8723801
TORQUE ROD ASSEMBLY	—	—
SPRINGS, MAIN	8407254	8758893
SPRINGS, AUXILIARY	NONE	NONE
TUBE ASSEMBLY FRONT AXLE	8742419	8742419
TUBE ASSEMBLY REAR AXLE	8742419	8742419
HUB & DRUM ASSEMBLY L.H.	—	—
HUB & DRUM ASSEMBLY R.H	—	—
HUB	8719915	8719915
DRUM	7411425	7411425
ADAPTER	7413231	7413231
WHEEL BEARING (OUTER)	MS19081-112	MS19081-112
WHEEL BEARING (INNER)	MS19081-112	MS19081-112
RELATED PARTS AND DATA		
WHEELS, P/N	MS53044-5	MS53044-5
WHEELS, SIZE	20X7.5	20X7.5
WHEELS, OFFSET	6 3/16	6 3/16
WHEELS, SINGLE OR DUAL	SINGLE	SINGLE
TIRES	MS35388-16	MS35388-16
TIRES, SIZE	9.00X20	9.00X20
TUBES	MS35392-13	MS35392-13
BRAKE SIZE	15X3	15X3
TRACK WIDTH	80	80
WHEEL BASE	50 1/2	50 1/2
SPRING CENTERS	55	55

MS35866-C1 INACTIVE FOR NEW DESIGN AFTER 12 JULY 1963. AUTHORIZED FOR PROCUREMENT, ONLY TO SUPPORT IN-SERVICE MATERIAL.

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TYPE 'D'

MS PART NO.	MS35866-DI
SUSPENSION INSTALLATION (TANDEM)	8336799
LOAD BEARING CAPACITY	50000 (LBS)
BOGIE ASSEMBLY	8336799
FRONT AXLE	8687126
REAR AXLE	8687127
TRUNNION ASSEMBLY (PIVOT AXLE)	8336792
BRAKE ASSEMBLY L.H. FRONT & R.H. REAR	8687077
BRAKE ASSEMBLY R.H. FRONT & L.H. REAR	8687078
TORQUE ROD ASSEMBLY	NONE
WALKING BEAM	11597645
SPRINGS, AUXILIARY	NONE
TUBE ASSEMBLY FRONT AXLE	8336982
TUBE ASSEMBLY REAR AXLE	8336982
HUB ASSEMBLY L.H.	10944309-1
HUB ASSEMBLY R.H.	10944309-2
HUB	8738111
DRUM ASSEMBLY	8687065
ADAPTER	—
WHEEL BEARING (OUTER)	MS19081-180
WHEEL BEARING (INNER)	MS 19081-134
RELATED PARTS AND DATA	
WHEEL, P/N	MS 53044-1
WHEEL, SIZE	15X7.5
WHEEL, OFFSET	6 1/8
WHEEL, SINGLE OR DUAL	DUAL
TIRE, P/N	MS 35390-5
TIRE SIZE	11.00X15
TUBES	MS35392-26
BRAKE SIZE	12 1/4 X 5 1/2
TRACK WIDTH	82 7/16
WHEEL BASE	42
WALKING BEAM CENTERS	49 1/8

APPROVED 27 FEBRUARY 1958 REVISED

P.A. Other Cust.	TITLE: AXLE ASSEMBLY, VEHICULAR NON DRIVING, TANDEM FOR TACTICAL WHEELED VEHICLES	MILITARY STANDARD MS 35866
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 7 OF 8

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P.A.
Other Cust

PROCUREMENT SPECIFICATION

SUPERSEDES:

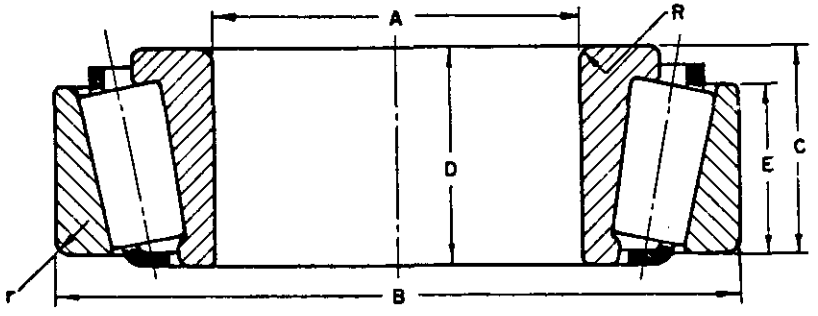
TITLE

AXLE ASSEMBLY, VEHICULAR NON DRIVING, TANDEM FOR TACTICAL WHEELED VEHICLES

MILITARY STANDARD

MS 35866

SHEET 8 OF 8



AXLE BEARING DATA

REFERENCE MS NUMBER	REF. MILITARY PART NUMBER: ASSEMBLY	CONE	CUP	A: INSIDE DIA.	B: OUTSIDE DIA.	C: BEARING WIDTH	D: CONE WIDTH	E: CUP WIDTH	R: CONE RADIUS	r: CUP RADIUS
MS19081-112	703116	705322	706771	2.6250	4.4375	1.1875	1.1830	.9375	.14	.13
MS19081-181	703077	705155	706691	3.3750	6.0000	1.5625	1.4300	1.1875	.14	.13
MS19081-134	703085	705187	706704	3.3475	5.9090	1.7500	1.8375	1.4375	.14	.13
MS19081-180	—	705167	707081	2.7500	5.1250	1.6250	1.6250	1.2500	.14	.13
MS19081-182	712288	712286	706691	3.5000	6.0000	1.5625	1.4300	1.1875	.25	.13
MS19081-138	703086	705194	706707	3.5000	6.3750	1.8750	1.9000	1.5000	.14	.13

NON DRIVING AXLE-CAPACITY (LBS)

REFERENCE MS NUMBER	2500 OUTER	2500 INNER	5000 OUTER	5000 INNER	10000 OUTER	10000 INNER	14000 OUTER	14000 INNER	18000 OUTER	18000 INNER	22000 OUTER	22000 INNER	25000 OUTER	25000 INNER
MS19081-112					✓	✓								
MS19081-181									✓					
MS19081-134											✓			✓
MS19081-180													✓	
MS19081-182										✓				
MS19081-138												✓		

NOTE: BEARING ASSEMBLIES MUST HAVE CUPS AND CONES MATCHED BY PART NUMBER AND MANUFACTURER.

FED. SUP. CLASS 2530

APPROVED 27 FEBRUARY 1958 REVISED

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